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Joe DiGiovanni

Data Scientist

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SKILLS

AI & Machine Learning: LLMs (RAG/Agents), NLP, PyTorch, XGBoost, Computer Vision, Scikit-learn, Causal Inference
Languages & Frameworks: Python, SQL, R, JavaScript, FastAPI, Next.js, React
Data Visualization: Streamlit, Tableau, Matplotlib, Seaborn, ggplot2
Data Engineering & Cloud: ETL Pipelines, REST APIs, DuckDB, Docker, GitHub Actions, GCP (Cloud Run), AWS
Core Competencies: Statistical Modeling, Predictive Analytics, Signal Isolation, ML Pipeline Development

EDUCATION

Master of Science, Applied Data Science, *The University of Chicago*
Bachelor of Science, Applied Mathematics, *University of California, Santa Barbara*

TECHNICAL EXPERIENCE

Founder & Lead Data Scientist

Sep 2025 — Present

FullCount AI

Remote

- Identified a structural gap in sports analytics tooling and built a public-facing platform that enables non-technical users to query complex Statcast data through natural language and receive modular, interactive visualizations without sacrificing analytical rigor.
- Diagnosed a core reliability failure in LLM-driven data querying of hallucinated SQL results and re-architected the system around a deterministic DuckDB routing layer, establishing correctness as a hard constraint rather than a best-effort outcome.
- Conducted structured discovery with early users and alpha testers to identify trust and consistency failures in existing AI sports tools, using those findings to drive the decision to replace raw LLM output with modular UI components by making results verifiable and repeatable rather than conversational and opaque.

AI Pipeline Engineer

Mar 2025 — Aug 2025

PowerKiosk, via UChicago Industry Capstone

Remote

- Led end-to-end system design for a 4-person team, owning technical decision-making throughout a six-month build; delivered a solution that earned Best in Show across the UChicago cohort.
- Transformed a corpus of 1,000+ heterogeneous financial and legal documents with no consistent structure by designing a multi-stage pipeline combining OCR, LLM inference, and automated rate calculations, making a previously unqueryable corpus analytically accessible.
- Replaced a manual document review process, measured in hours per query, with a RAG-powered natural language interface that returned results in seconds, eliminating a bottleneck that had no viable manual solution at scale.

Analytics Research Assistant

Sept 2023 — June 2024

UCSB Baseball, University of California, Santa Barbara (NCAA D1)

Santa Barbara, CA

- Applied predictive modeling in a domain defined by small samples and high variance to 750+ player entities with limited signal by developing an evaluation framework that identified high-ROI targets, isolating meaningful performance indicators from noise and directly shaping selection strategy.
- Built Python ETL pipelines to analyze pitcher Vertical Approach Angles and release point consistency using Trackman radar telemetry, translating raw tracking data into actionable metrics for coaching staff
- Integrated unstructured video and radar tracking data to produce scouting analyses on opposing teams, synthesizing multimodal signals into a single defensible picture of opponent tendencies in high-leverage situations.
- Modeled team bunting tendencies through Expected Value analysis to quantify run-scoring impact across game situations, translating a coaching intuition into a data-backed framework that sharpened in-game offensive decision-making.

PROJECTS

Independent Quantitative Research on MLB Pitch Tunneling

Jun 2025 — Present

The University of Chicago

Remote

- Partnered with the University of Chicago to lead an end-to-end independent research project on a domain problem with no established feature framework and built one from first principles, defining measurable constructs where none existed to achieve a 150%+ improvement in model R^2 while preserving interpretability and domain validity.
- Applied causal inference techniques to identify the primary mechanism driving observed outcomes, demonstrating a measurable increase (up to +23.4%) in target behavioral responses within strategically defined regions of the feature space
- Created interpretable visualizations translating complex statistical outputs from research experiments into clear, actionable insights for non-technical stakeholders while preparing findings for submission to a peer-reviewed journal.